

ARPAN SANDHU

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PROFESSIONAL SUMMARY

Data-driven Computer Engineering student (Thapar Institute, Batch 2027, CGPA 8.63) with hands-on experience in business analytics, machine learning, and BI dashboard development. Proven ability to translate complex datasets into strategic, actionable insights that support data-driven decision-making across cross-functional stakeholders. Proficient in Python, SQL, Microsoft Excel, Power BI, and Tableau. Experienced in EDA, regression analysis, hypothesis testing, and segmentation analysis. Eager to apply analytical and problem-solving capabilities in a fast-paced SaaS or analytics environment.

EXPERIENCE

Machine Learning Intern — Staqa (Remote)

June 2026 – Present

- Accelerated model delivery by 30%, measured by sprint velocity metrics, by designing and automating end-to-end data preprocessing and EDA pipelines in Python, reducing manual analyst effort across three concurrent business use cases.
- Improved cross-team handoff efficiency by 40%, measured by onboarding time reduction, by producing structured technical documentation and workflow optimisation guides aligned with Agile sprint cycles, enabling faster knowledge transfer for incoming team members.
- Supported data-driven decision-making by translating AI/ML model outputs into clear business recommendations for non-technical stakeholders, collaborating with senior engineers via Git in a fast-paced global SaaS environment.
- Reduced cloud infrastructure costs by automating data ingestion and model training pipelines on AWS using Unix shell scripting, cutting manual pipeline effort by an estimated 50% per cycle.

Research Intern — Thapar Institute of Engineering and Technology, Patiala

May 2025 – Present

- Improved threat identification accuracy by 68%, measured against baseline classification benchmarks, by developing a behaviour-based anomaly detection framework using clustering and supervised learning on real-time user interaction data across 50,000 daily sessions.
- Reduced potential threat response time by 45%, measured by alert latency benchmarks, by engineering a real-time feature extraction pipeline that processed behavioural signals and fed outputs to a classification model integrated into a cybersecurity monitoring platform.
- Enhanced stakeholder decision-making by delivering documented technical findings, visualisation dashboards, and architectural recommendations to faculty, translating complex ML methodology into actionable risk-mitigation strategies.
- Maintained zero merge conflicts over six months, measured across 4-person research team, by coordinating Git branching strategies and Agile sprint planning, ensuring consistent delivery of research milestones.

PROJECTS

ML Model Serving Platform | Personal Project | June 2025 – Aug 2025

- Reduced model deployment downtime to zero, measured across three concurrent production model versions, by implementing A/B testing, blue-green deployment, and model versioning using FastAPI and Docker on AWS-hosted cloud architecture.
- Improved database query performance by 35%, measured by average query response time, by designing a normalised PostgreSQL schema for model metadata and prediction logs, applying indexing and partitioning strategies aligned with data modelling best practices.
- Created analytical BI dashboards tracking model performance KPIs, measured by model promotion cycle time reduced by 35%, by integrating REST API endpoints with a React.js front-end and Redis caching handling 500 requests per minute.

AI Intrusion Detection System | Personal Project | Jan 2025 – May 2025

- Achieved 94% classification accuracy and 99% ROC-AUC, measured on a holdout test set, by training an XGBoost model on the UNSW-NB15 dataset (257K+ records) with SMOTE-based class balancing and iterative feature engineering, reducing false positives by 22%.
- Reduced analyst response time by 50%, measured by time-to-alert benchmarks, by building a real-time Streamlit BI dashboard and Flask REST API for live inference, enabling non-technical stakeholders to monitor network anomalies without SQL access.

- Automated data ingestion and model retraining, measured by cutting manual pipeline effort from three hours to under 15 minutes per cycle, by writing Unix shell scripts and cron jobs on a Linux server for continuous model updates.

Movie Recommendation System | Personal Project | Jan 2024 – June 2024

- Improved recommendation relevance to precision@10 of 0.73, measured against collaborative filtering baselines, by building a hybrid recommender combining Pearson correlation and KNN on the MovieLens dataset.
- Enabled real-time personalised suggestions for 1,000 test users, measured by system response latency, by delivering a full-stack Flask and JavaScript web interface integrated with a Node.js-compatible backend.

TECHNICAL SKILLS

Analytics & Business Intelligence: Microsoft Excel (advanced), SQL, Power BI, Tableau (exposure), Streamlit, EDA, Segmentation Analysis, Data Visualisation, Regression Analysis, Hypothesis Testing

Data Science & Python: Python, Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, Matplotlib, Seaborn, Jupyter Notebook

Cloud & Infrastructure: AWS (EC2, S3, Lambda), Azure, Docker, Linux, Unix, Bash Scripting

Web & APIs: FastAPI, Flask, REST APIs, React.js, Node.js

Databases: PostgreSQL, MySQL, Data Modelling, Indexing

Tools & Workflow: Git, GitHub, Agile, Scrum, Waterfall, Technical Documentation, Stakeholder Reporting

EDUCATION

Thapar Institute of Engineering and Technology, Patiala

B.E. Computer Engineering | CGPA: 8.63 | July 2023 – May 2027

Dashmesh Public School, Faridkot, Punjab

Senior Secondary (PCM), CBSE | 95.4% | April 2021 – May 2023

St. Mary's Convent School

Secondary, ICSE | 98% | March 2021